



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to KISHORE FUEL STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

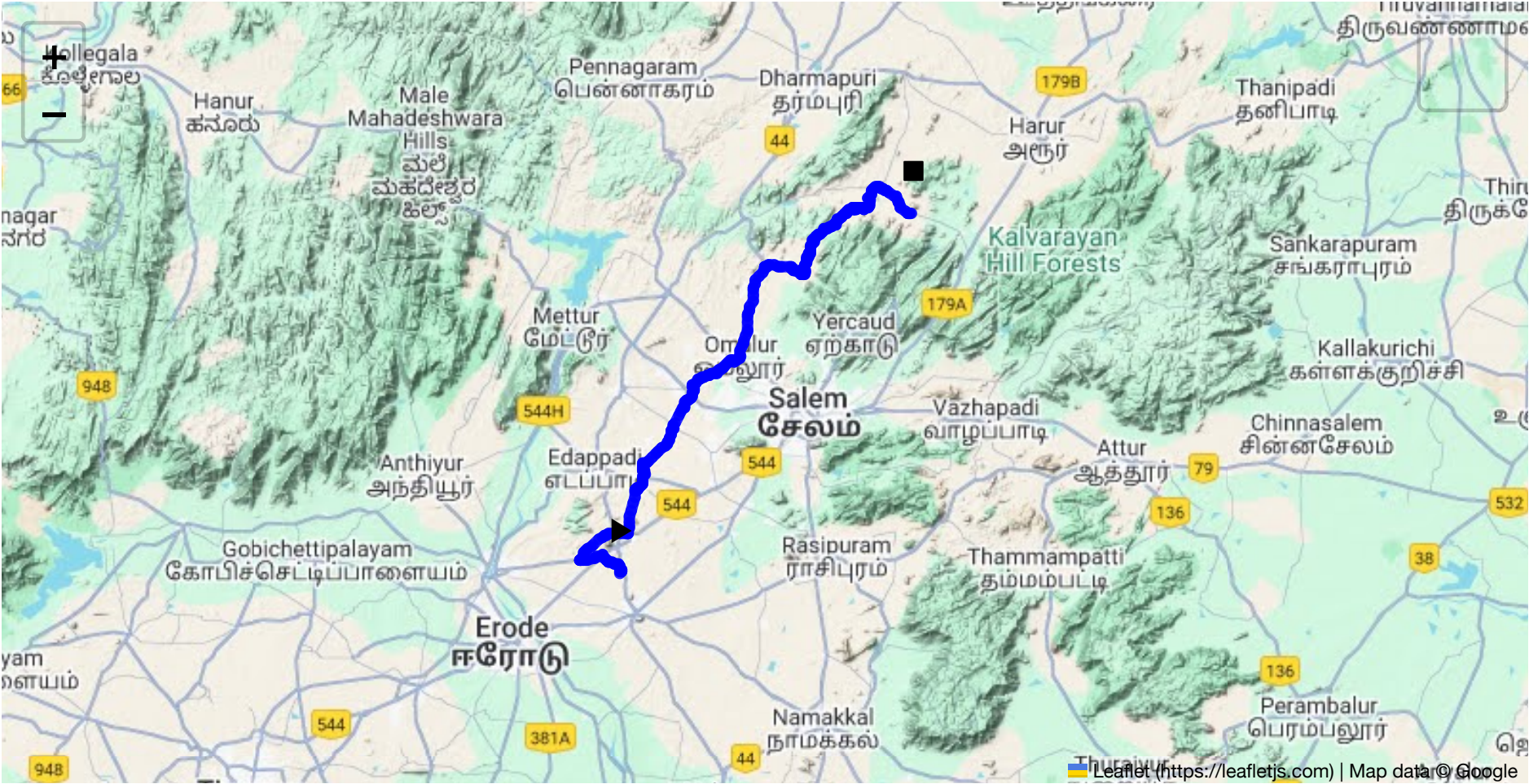
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 104.59 km
Estimated Duration: 2.5 hours
Adjusted Duration (Heavy Vehicle): 3.2 hours

Start: (11.4381, 77.8734)
End: (11.944429, 78.292952)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route spans approximately 104.59 kilometers through southern Tamil Nadu, beginning at Sangagiri and ending at Pallipatti. It passes through several key points, including Bhavani Main Road, SH 86 in Ayveli, and Salem, terminating near Omalur before reaching its final destination. The route consists of a mix of highways and urban roads, with varying levels of traffic and infrastructure quality.

2. Weather Conditions and Potential Hazards

Tamil Nadu tends to have a tropical climate, with high temperatures and humidity levels. The area can experience heavy rainfall during the northeast monsoon from October to December, which may result in waterlogging and reduced visibility, particularly on poorly drained roads. Dust storms and high temperatures in summer could also pose challenges.

3. Traffic Patterns and Congestion-Prone Areas

- Peak Hours:** Expect morning and evening congestion in urban centers like Salem, usually from 8:00 to 10:00 AM and 5:00 to 7:00 PM.
- Congestion-Prone Areas:** Bhavani Main Road and approaches to Salem are often congested. Entry and exit points of Omalur may also experience slow traffic, especially due to commercial activities around the Triveni Car Showroom.

4. Road Quality and Infrastructure

The route includes a mix of well-paved highways and some rural roads. The highways are generally maintained, but watch for occasional potholes or roadworks, particularly in less urbanized areas. Signage quality may vary, impacting navigation especially in rural segments.

5. Suggestions for Alternative Routes

In case of emergencies, consider alternate routes bypassing Salem via rural roads if congestion or incidents occur. Local routes could provide detours, although these might be less direct and slower.

6. Local Regulations on Hazardous Material Transport

Tamil Nadu enforces strict regulation on the transport of hazardous materials; adherence to designated routes and timing (often avoiding peak hours) is crucial. Permits and clear labeling should be ensured.

7. Historical Incidents

There have been occasional incidents involving heavy vehicles on this route, typically due to road conditions or congestion-related collisions, especially in Salem. Awareness and adherence to speed limits can mitigate risks.

8. Environmental Considerations

The route traverses some agricultural areas; any spillage of hazardous materials could impact local ecosystems. Maintaining precautions near water bodies, such as rivers or tanks, which are vital for local agriculture, is crucial.

9. Communication Coverage

Most urban and semi-urban segments have reliable mobile coverage. Potential dead zones exist in rural sections, particularly between Ayveli and Sangagiri. Always have a secondary communication means such as two-way radio.

10. Emergency Response Times

Emergency services in urban centers like Salem can respond relatively quickly (approximately 15-30 minutes). In rural areas, response times may extend up to an hour or more due to longer travel distances and road conditions.

11. Overall Summary of Risk Assessment

The route presents moderate risks primarily due to traffic congestion, weather conditions during monsoon season, and variable road quality. Preparedness for emergency detours and communication issues are important to mitigate risks. Practicing safe driving, especially during peak hours and poor weather, along with awareness of environmental sensitivities, is essential for safe transportation of hazardous materials. Adherence to regulations and seeking updated local advisories are advised for the latest information.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45353, 77.85697	15 KM/Hr
4	Turn	Medium	11.45781, 77.85203	30 KM/Hr
5	Turn	High	11.46307, 77.84941	15 KM/Hr
6	Turn	High	11.45542, 77.81725	15 KM/Hr
7	Turn	High	11.45636, 77.81678	15 KM/Hr
8	Turn	Medium	11.49383, 77.88501	30 KM/Hr
9	Blind Spot	Blind Spot	11.49400, 77.88538	10 KM/Hr
10	Turn	High	11.49469, 77.88488	15 KM/Hr
11	Turn	High	11.55618, 77.89789	15 KM/Hr
12	Turn	Medium	11.55629, 77.89888	30 KM/Hr
13	Turn	High	11.58228, 77.90841	15 KM/Hr
14	Turn	High	11.58340, 77.90637	15 KM/Hr
15	Turn	Medium	11.59201, 77.90649	30 KM/Hr
16	Turn	Medium	11.63832, 77.94818	30 KM/Hr
17	Turn	Medium	11.64161, 77.95177	30 KM/Hr
18	Turn	Medium	11.64226, 77.95200	30 KM/Hr
19	Turn	Medium	11.64716, 77.95118	30 KM/Hr
20	Turn	Medium	11.67868, 77.96836	30 KM/Hr
21	Turn	Medium	11.70077, 77.97957	30 KM/Hr
22	Turn	High	11.70223, 77.97860	15 KM/Hr
23	Turn	Medium	11.71402, 77.99586	30 KM/Hr
24	Turn	Medium	11.71918, 78.01127	30 KM/Hr
25	Turn	Medium	11.73769, 78.03451	30 KM/Hr
26	Turn	Medium	11.73785, 78.03551	30 KM/Hr
27	Turn	High	11.73651, 78.03871	15 KM/Hr
28	Turn	Medium	11.74271, 78.04717	30 KM/Hr
29	Turn	Medium	11.86763, 78.08336	30 KM/Hr
30	Turn	Medium	11.87000, 78.08385	30 KM/Hr
31	Turn	Medium	11.87075, 78.08366	30 KM/Hr
32	Blind Spot	Blind Spot	11.87109, 78.08373	10 KM/Hr
33	Turn	Medium	11.87089, 78.08395	30 KM/Hr
34	Turn	Medium	11.87077, 78.08526	30 KM/Hr
35	Turn	Medium	11.87257, 78.08752	30 KM/Hr
36	Turn	Medium	11.86901, 78.12052	30 KM/Hr
37	Turn	Medium	11.86853, 78.12149	30 KM/Hr
38	Turn	High	11.85820, 78.13580	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
39	Turn	Medium	11.86071, 78.13960	30 KM/Hr
40	Turn	Medium	11.86394, 78.14013	30 KM/Hr
41	Turn	Medium	11.86670, 78.13878	30 KM/Hr
42	Turn	Medium	11.88446, 78.14472	30 KM/Hr
43	Turn	Medium	11.88522, 78.14465	30 KM/Hr
44	Turn	Medium	11.88752, 78.14563	30 KM/Hr
45	Turn	Medium	11.90568, 78.15290	30 KM/Hr
46	Turn	High	11.90785, 78.15209	15 KM/Hr
47	Turn	Medium	11.91215, 78.15588	30 KM/Hr
48	Turn	Medium	11.91305, 78.15611	30 KM/Hr
49	Turn	Medium	11.92454, 78.17106	30 KM/Hr
50	Turn	Medium	11.92819, 78.18025	30 KM/Hr
51	Turn	Medium	11.92803, 78.18203	30 KM/Hr
52	Turn	Medium	11.92928, 78.18365	30 KM/Hr
53	Turn	Medium	11.93550, 78.18702	30 KM/Hr
54	Turn	Medium	11.95110, 78.20634	30 KM/Hr
55	Turn	Medium	11.95236, 78.20766	30 KM/Hr
56	Turn	High	11.95258, 78.20832	15 KM/Hr
57	Turn	Medium	11.95173, 78.21009	30 KM/Hr
58	Turn	High	11.95221, 78.21163	15 KM/Hr
59	Turn	High	11.95162, 78.21312	15 KM/Hr
60	Turn	Medium	11.95237, 78.21447	30 KM/Hr
61	Turn	High	11.95218, 78.21514	15 KM/Hr
62	Turn	Medium	11.95274, 78.21574	30 KM/Hr
63	Turn	Medium	11.95250, 78.22509	30 KM/Hr
64	Turn	High	11.95178, 78.22596	15 KM/Hr
65	Turn	Medium	11.96176, 78.23443	30 KM/Hr
66	Turn	Medium	11.96542, 78.23289	30 KM/Hr
67	Turn	High	11.96664, 78.23287	15 KM/Hr
68	Blind Spot	Blind Spot	11.96695, 78.23203	10 KM/Hr
69	Turn	High	11.97321, 78.23713	15 KM/Hr
70	Turn	High	11.97761, 78.23553	15 KM/Hr
71	Turn	High	11.98244, 78.24307	15 KM/Hr
72	Turn	High	11.98207, 78.24331	15 KM/Hr
73	Turn	High	11.98346, 78.24617	15 KM/Hr
74	Turn	Medium	11.96850, 78.27106	30 KM/Hr
75	Turn	Medium	11.94553, 78.28890	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
8	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
10	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
11	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
12	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
13	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
14	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
15	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
16	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
17	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
18	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
19	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
20	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
21	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
22	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
24	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
26	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
27	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
28	clinic	Kamala Clinic	11.8723673, 78.0874195	30 km/h	Medium
29	hospital	Kamala Hospital	11.8722588, 78.0899189	30 km/h	Medium
30	hospital	GJB Hospital	11.8713998, 78.0980288	30 km/h	Medium
31	clinic	Pranav Clinic	11.8712846, 78.0998167	30 km/h	Medium
32	hospital	Bethel Hospital	11.8699472, 78.1388598	30 km/h	Medium
33	hospital	Amirtham Hospital	11.981994, 78.248038	30 km/h	Medium
34	hospital	Grs Hospital	11.9824068, 78.2481763	30 km/h	Medium
35	hospital	Raja Hospital, Dharmapuri	11.9792908, 78.2509482	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
9	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
23	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
25	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45636, 77.81678



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49383, 77.88501



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49400, 77.88538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55618, 77.89789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58340, 77.90637



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59201, 77.90649



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63832, 77.94818



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64161, 77.95177



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64226, 77.95200



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64716, 77.95118



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67868, 77.96836



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70077, 77.97957



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70223, 77.97860



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71402, 77.99586



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71918, 78.01127



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73769, 78.03451



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73785, 78.03551



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03871



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74271, 78.04717



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86763, 78.08336



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87000, 78.08385



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87075, 78.08366



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.87109, 78.08373



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87089, 78.08395



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87077, 78.08526



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87257, 78.08752



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86901, 78.12052



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95110, 78.20634



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.97321, 78.23713



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.97761, 78.23553



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.98244, 78.24307



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.98207, 78.24331



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.98346, 78.24617



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96850, 78.27106



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.94553, 78.28890

